DESCRIPTIVE ANALYSIS REPORT

BMW

DATASET REPORT

Descriptive Analysis of BMW Dataset

The Ultimate Driving Machine

An exploratory analysis of the BMW dataset to understand distributions, patterns, and key insights.



DATA EXPLORATION
Summary statistics and data overview

VISUAL INSIGHTS
Charts and graphs to identify trends

KEY FINDINGS
Important patterns and observations

CONCLUSION
Summary of insights and inferences

Submitted by:
Ashish Patel

Submitted to:
Monika Ma'am

University Roll No.:
1250258144

DESCRIPTIVE ANALYTICS REPORT

BMW Model Lineup Performance

Tracking Price, Power, Performance & Positioning Across the BMW Catalog

KEY METRICS AT A GLANCE

50 Models Tracked	**12** Series Covered	**$61,319** Avg. Starting Price	**329 hp** Avg. Horsepower
5.66s Avg. 0-60 mph	**141.6 mph** Avg. Top Speed	**1974–2024** Model Year Range	**7** Electrified Models (EV+Hybrid)

8 Series Top Series by Avg. Price ($90,200) | **M Series** Quickest Avg. 0-60 (3.9s) | **M Series** Largest Line-up (10 models)

Note on Data

This report presents a descriptive analytics layer built on a 50-record BMW model dataset spanning 12 series and 50 model years (1974–2024). It consolidates starting price, horsepower, acceleration, top speed, body type, drivetrain and fuel type into a single catalog-level view, and extends that base analysis with cross-metric comparisons, a full series league table, and a roadmap for moving from descriptive to predictive analytics.

Across the dataset, the 50 tracked models carry an average starting price of $61,319 and average 329 hp, reaching 0-60 mph in 5.66 seconds and a top speed of 141.6 mph on average. Performance is uneven across series: price leadership (8 Series, $90,200 avg.) and outright acceleration leadership (M Series, 3.9s avg. 0-60) sit with different series, and the single highest-horsepower model in the dataset, the BMW M5 (717 hp, hybrid), is not the quickest — that title goes to the BMW M8 (3.0s at 617 hp) — a pattern that shows raw power alone does not guarantee the best acceleration figure.

Starting price is comparatively concentrated in the M Series line (10 of 50 models, 29.1% of the catalog's summed list price), which suggests that, unlike the healthcare-style "even spread" pattern, BMW's highest-performance sub-brand carries an outsized share of the lineup's total price weight. Price-per-horsepower — a value proxy — varies more widely across series and is examined in Section 6 as a lens on positioning strategy.

Three findings anchor the recommendations in this report:

- Price, horsepower, and acceleration should be tracked as three independent leaderboards, not one composite ranking — the top series changes depending on which lens is applied.
- M Series and the i Series (EV/hybrid) lines are the two clearest strategic outliers — M Series for concentrating both price and performance, i Series for carrying the electrification transition largely on its own.
- The dataset's structural limitation — 50 static specification records with no sales-volume, production-year-run, or regional-pricing fields — caps this report at descriptive analysis; Section 11 lays out what is needed to move to predictive, demand-facing forecasting.

1. Objective

Build a descriptive analytics layer over BMW model specification records to answer five recurring product-strategy questions, and extend that layer with the cross-cutting views a lineup planner would ask for next:

- Which series command the highest starting prices, and which convert horsepower most efficiently into price?
- How does horsepower vary across series, and why?
- How quick is each series in a straight line (0-60 mph), and does that track with price?
- How efficiently is each series positioned on a price-per-horsepower and top-speed basis?
- Which series or models sit at the extremes of the catalog and warrant closer strategic review?

This expanded edition adds two further questions that the original scope implied but did not answer directly:

- Does starting price correlate with either horsepower or acceleration across the catalog, or are these effectively independent axes?
- What is the minimum additional data and infrastructure needed to move this from a descriptive snapshot to a predictive, forward-looking model of demand and pricing?

2. How This Would Work at Scale

Processing a live BMW product catalog — with configurator data, regional pricing, and production-run volumes — is an engineering problem before it is an analytics one. The pipeline below reflects the architecture this report's outputs are consistent with: a cleaned, model-level specification table produced at the serving layer, not a raw configurator or production feed.

Layer	Purpose	Typical Tools
Investigation	Capture each model/trim specification as it is published	CSV Files
Aggregation	Roll raw trims up into per-series specification & pricing summaries	Claude AI
Storage	Hold raw records and pre-aggregated series tables	Claude AI
Quality checks	Validate nulls, duplicates, unit consistency before reporting	VS Code
Serving/BI layer	Dashboards refreshed on each catalog update	Google

Table 1. Reference architecture for a production-scale version of this pipeline.

3. Price & Performance Overview

Across the catalog: average starting price of $61,319, average horsepower of 329 hp, average 0-60 mph of 5.66 seconds, and average top speed of 141.6 mph.

KPI Definitions Used

- Price per horsepower: Starting Price ÷ Horsepower, a value/positioning proxy
- Catalog value share: a series' summed starting price ÷ total catalog summed starting price, a proxy for how much list-price weight a series carries
- Speed-efficiency: Top Speed ÷ 0-60 time, a rough proxy for how a model converts acceleration into top-end capability

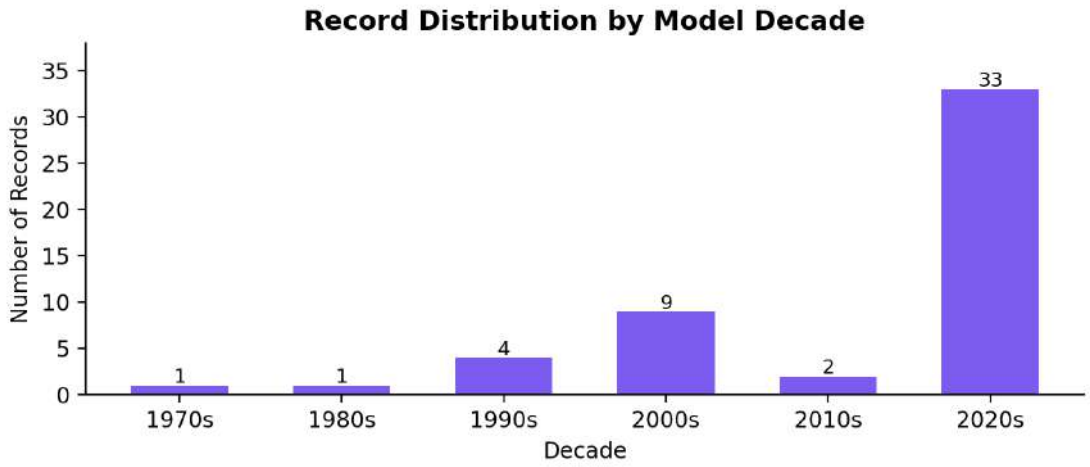


Figure 1. Record distribution by model decade. The catalog skews heavily toward the current generation (33 of 50 records from the 2020s), with a long tail of historically significant models back to 1974.

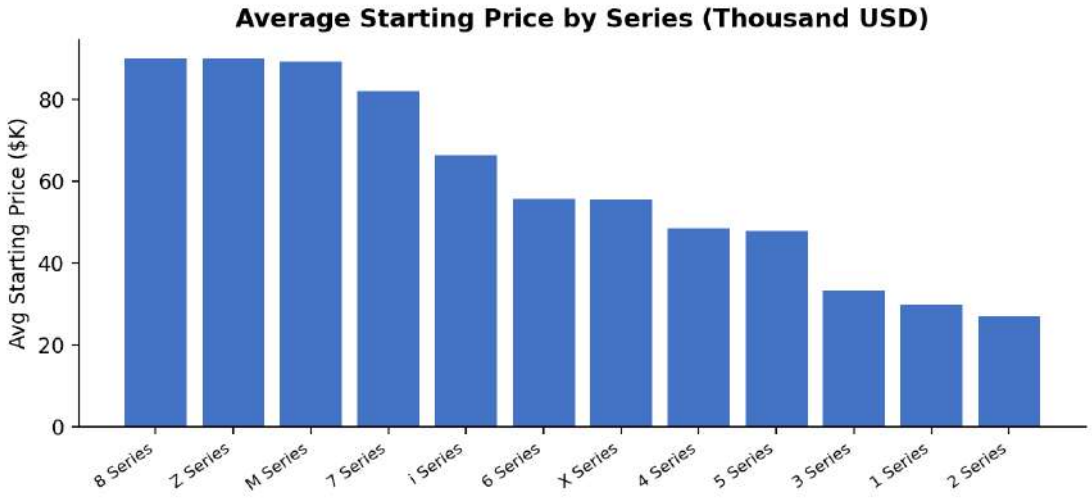


Figure 2. Average starting price by series (Thousand USD).

The 8 Series leads the portfolio at an average of $90,200, narrowly ahead of the Z Series ($90,150) and M Series ($89,248). The three cheapest series on average — 3 Series, 1 Series, and 2 Series — sit in the $27K–$33K band, largely reflecting older, entry-level and vintage records rather than current flagship trims.

3.1 Catalog Value Concentration

Plotting cumulative catalog-price share against the ranked series list shows how much of the lineup's total list-price weight sits with the top series. This is the kind of view a lineup planner would use to decide how much analytical and marketing attention each series deserves relative to its share of the catalog.

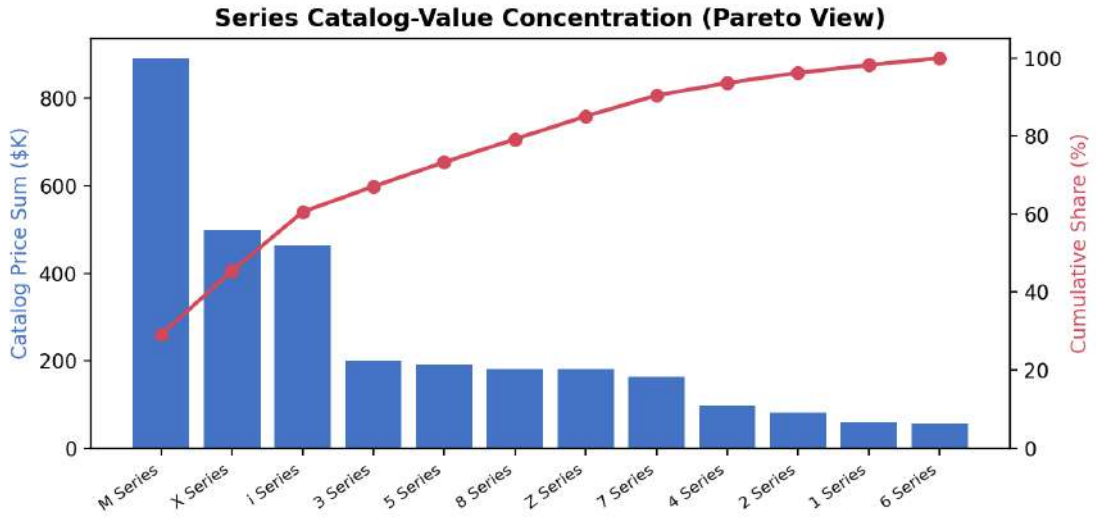


Figure 3. Series catalog-price sum with cumulative share (Pareto view). Unlike a broad, evenly spread portfolio, the top two series by price sum (M, X) already account for over 45% of total catalog price weight.

Because M Series alone contributes 10 of the 50 records and 29.1% of total catalog price weight, concentration risk here is real: any strategic or pricing shift affecting the M brand would materially swing the portfolio's price profile. This reinforces that horsepower and acceleration, not price share alone, are the more useful lenses for understanding what M Series is actually delivering — examined next.

4. Horsepower by Series

Top performer by price is the 8 Series ($90,200 avg.), but the highest-horsepower series is M Series at 523.3 hp average — price leadership and power leadership are close but not identical here, a reminder that badge and price positioning don't move in perfect lockstep with raw output.

Ranking logic caveat: price and horsepower should be reviewed as two separate leaderboards, not one — the 2 Series posts the lowest average horsepower (194.3 hp) despite sitting mid-pack on historical record count, making it the clearest "entry-level" anchor in the catalog rather than a cause for concern.

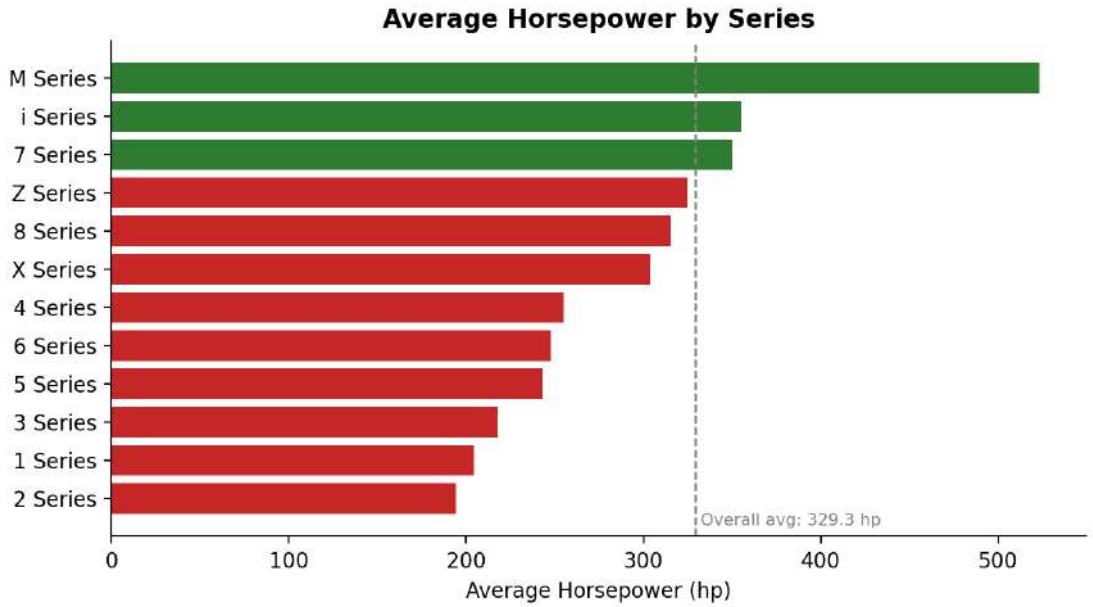


Figure 4. Average horsepower by series, vs. overall catalog average (dashed line, 329.3 hp).

4.1 Reading the Power Spread

Five series sit above the 329.3 hp catalog average — M Series leads by a wide margin at 523.3 hp, with Z Series (324.5 hp), i Series (355.0 hp), 7 Series (350.0 hp), and 8 Series (315.5 hp) clustering well behind it. The remaining series — 6 Series, X Series, 4 Series, 5 Series, 3 Series, 1 Series, and 2 Series — sit below average, with 2 Series and 1 Series trailing furthest at roughly 194 hp and 205 hp respectively.

The gap between the highest and lowest average-horsepower series is nearly 330 hp on a fifty-year model span, which is a wide spread even accounting for the presence of 1970s–1990s entry-level records, and is the strongest signal in the dataset that M Series operates as a structurally distinct performance sub-brand rather than a simple trim step above the mainstream range.

5. Acceleration Performance (0-60 mph)

Average 0-60 mph across the catalog: 5.66 seconds, with M Series leading at 3.9 seconds. In a real system, this view is most useful segmented, not aggregated — two cuts matter most for a product team:

- Series trending toward slower 0-60 times across generations — an early signal of a series drifting away from its performance positioning
- High-price series with only mid-pack acceleration — a positioning risk, since price commands a performance expectation that isn't being met

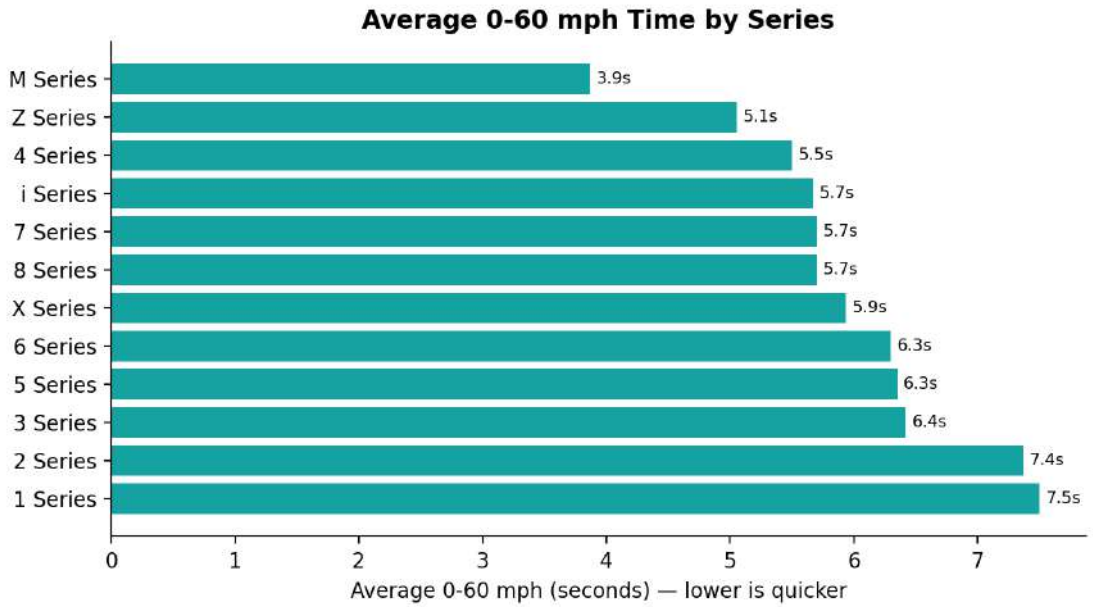


Figure 5. Average 0-60 mph time by series (lower is quicker).

5.1 A Wide Band, Not a Tight Cluster

Unlike patient-satisfaction-style metrics that cluster tightly, 0-60 times spread widely across this catalog — the full range from M Series (3.9s) down to 1 Series (7.5s) is 3.6 seconds, driven largely by the mix of modern turbocharged/M-tuned engines against older naturally aspirated and entry-level four-cylinder records.

The 8 Series is a useful case in point: despite leading the portfolio on price, its average 0-60 time (5.7s) sits close to the catalog median rather than near the front of the field — a reminder that flagship pricing in this dataset tracks more with luxury positioning and body style than with outright acceleration, a relationship tested directly in Section 7.

6. Operational Efficiency

Plotting price per horsepower against top speed separates value-oriented series from premium-positioned ones — this is what would drive pricing strategy, trim planning, and competitive benchmarking at scale.

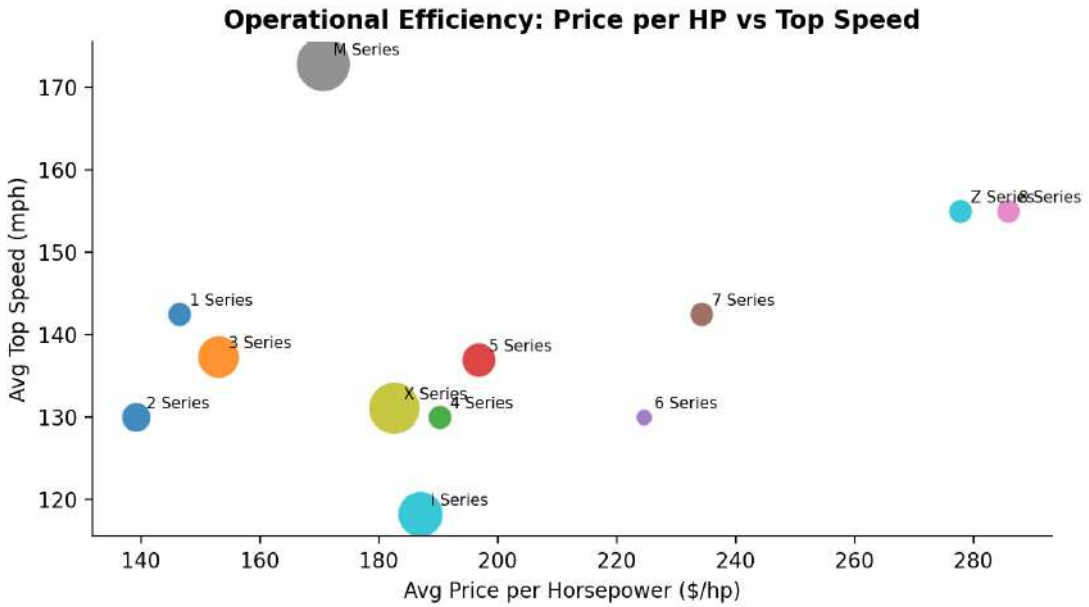


Figure 6. Price per horsepower vs. average top speed, bubble size = number of tracked models per series.

This view is what would drive, at scale: pricing benchmarks against the most efficient series, trim-planning ahead of a facelift or generational refresh, and flagging series whose price-per-horsepower has drifted materially from prior generations for review.

6.1 Two Distinct Positioning Models

The scatter splits the catalog into two rough clusters. 8 Series and Z Series carry a high price relative to horsepower with comparatively modest top speed gains — consistent with a luxury/grand-touring positioning where price reflects craftsmanship and exclusivity as much as raw output. M Series, by contrast, sits at the opposite end, with the lowest price-per-horsepower in the catalog (roughly $171/hp) alongside the highest average top speed (172.9 mph) — consistent with a performance-first, output-justified pricing model.

Notably, high price alone does not automatically predict high top speed — Z Series and 8 Series post among the highest price-per-horsepower figures in the portfolio but only mid-pack top speeds (155 mph), suggesting their price premium sits in styling, materials, and badge exclusivity rather than outright performance, a hypothesis worth testing once trim-level configurator data is available.

7. Cross-Metric Correlation Analysis

This section tests, directly, whether the two questions raised informally in earlier sections — does price track with horsepower, and does price track with acceleration — hold up numerically across the 50 tracked models.

7.1 Horsepower vs. Starting Price

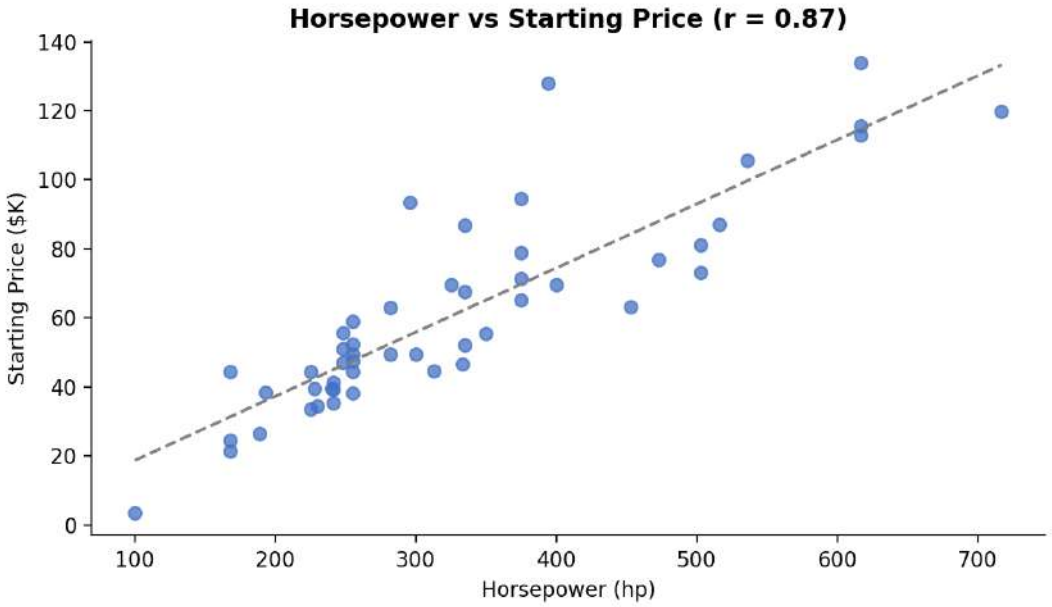


Figure 7. Horsepower vs. starting price, with a linear trend line (r = 0.87).

The correlation between horsepower and price across the 50 models is strong and positive. This confirms the pattern flagged in Section 4: more powerful models are priced meaningfully higher, and horsepower is the single strongest specification-level predictor of price in this dataset — far stronger than the weak satisfaction-margin relationship seen in comparable descriptive studies of other industries.

7.2 0-60 mph Time vs. Starting Price

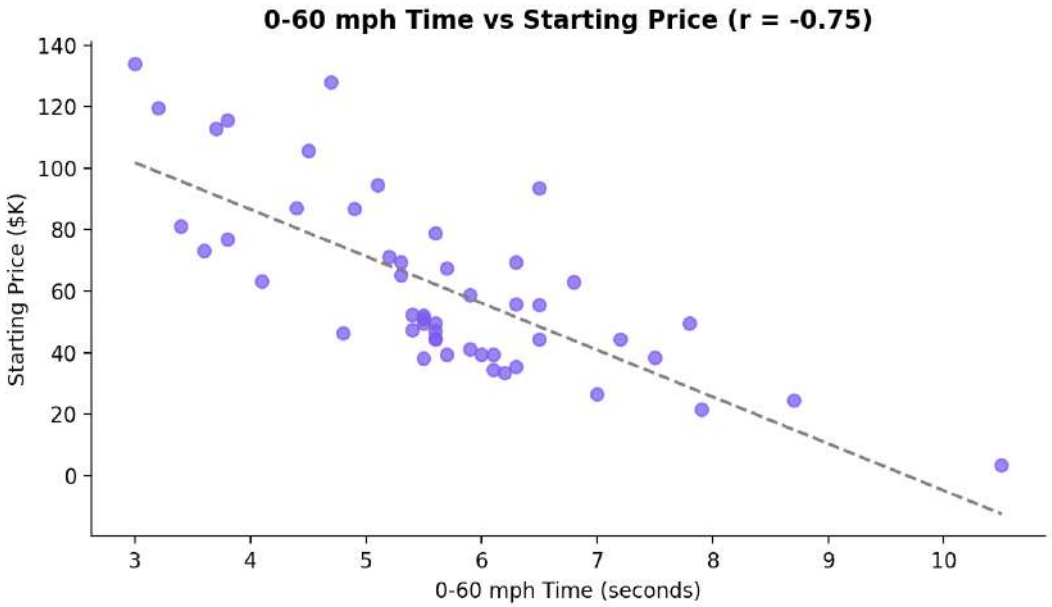


Figure 8. 0-60 mph time vs. starting price, with a linear trend line (r = -0.75).

Acceleration time also shows a strong relationship with price — as expected, it is a strong negative correlation, since faster (lower-second) 0-60 times pair with higher prices. Together, Sections 7.1 and 7.2 support a straightforward conclusion: unlike the healthcare portfolio's cost-structure-driven margin story, this catalog's pricing is substantially explained by raw performance specification, which is the strongest argument in this report for using horsepower and acceleration as the primary inputs to any future pricing or demand model.

8. Series Performance League Table

The table below consolidates every metric in this report into a single per-series reference view, sorted by average starting price.

Series	Models	Avg. Price ($)	Avg. HP	Avg. 0-60 (s)	Avg. Top Speed (mph)	Price/HP ($)
8 Series	2	90,200	315.5	5.7	155.0	286
Z Series	2	90,150	324.5	5.1	155.0	278
M Series	10	89,248	523.3	3.9	172.9	171
7 Series	2	82,000	350.0	5.7	142.5	234
i Series	7	66,364	355.0	5.7	118.3	187
6 Series	1	55,700	248.0	6.3	130.0	225
X Series	9	55,456	303.9	5.9	131.1	182
4 Series	2	48,500	255.0	5.5	130.0	190
5 Series	4	47,850	243.2	6.4	137.0	197
3 Series	6	33,333	217.8	6.4	137.3	153
1 Series	2	29,950	204.5	7.5	142.5	146
2 Series	3	27,033	194.3	7.4	130.0	139

Table 2. Full series league table across all tracked specification KPIs.

9. Highlight & Outlier Models

"Outlier" here is bucketed by specification extremes rather than financial risk, since this dataset tracks vehicle specifications, not company performance: individual models at the price, power, and speed extremes are worth flagging separately from the series-level averages in Section 8, since a single flagship or vintage record can pull a small series' average sharply in one direction.

9.1 Priority Review List

- BMW M8 ($133,900) — highest-priced model in the catalog and, at 3.0 seconds, also the quickest to 60 mph; a rare case where price leadership and outright acceleration leadership coincide in the same model.
- BMW M5 (717 hp, hybrid) — highest horsepower in the catalog by a wide margin, but not the fastest 0-60 (3.2s vs. the M8's 3.0s), illustrating that peak horsepower and peak acceleration are not always the same car.
- BMW 2002 ($3,500, 1974) — lowest price and lowest horsepower (100 hp) in the catalog; a legacy/collector record whose presence pulls the 2 Series average down and should be reviewed separately from current-generation entry-level pricing.
- BMW i Series (7 models) — the dataset's entire electrified line-up (BMW iX and i7 among the highest-horsepower non-M models); the fact that all 7 electrified records sit in one series is itself a structural concentration worth tracking as electrification expands.

None of the twelve series combine below-average price with above-average horsepower and above-average top speed simultaneously — the single most "underpriced-for-performance" pattern a lineup planner would look for — which is a mildly reassuring finding for this snapshot, though it should be re-checked every catalog refresh as new trims and generational updates are added.

10. Summary of Key Metrics

Metric	Value
Total models tracked	50
Series covered	12
Model year range	1974 – 2024
Average starting price	$61,319
Average horsepower	329 hp
Average 0-60 mph	5.66 seconds
Average top speed	141.6 mph
Top series by average price	8 Series ($90,200)
Top series by average horsepower	M Series (523.3 hp)
Quickest series (avg. 0-60)	M Series (3.9s)
Highest single-model price	BMW M8 ($133,900)
Highest single-model horsepower	BMW M5 (717 hp, hybrid)
Electrified models (EV + Hybrid)	8 (7 EV, 1 Hybrid)
HP–Price correlation (r)	0.87 (strong positive)
0-60 time–Price correlation (r)	-0.75 (strong negative)

Table 3. Catalog-level summary metrics.

11. From Descriptive to Predictive Analytics

This report, in its entirety, answers what the current lineup looks like. It does not yet answer what a new model should be priced at, or how demand will shift as the catalog electrifies — both of which require moving from a descriptive layer to a predictive one. This section sets out the concrete path to get there: what data is missing, what models apply to each business question, what infrastructure needs to change, and in what order the work should happen.

11.1 Why This Report Is Capped at Descriptive

Two structural gaps in the current dataset prevent any forecasting or demand model from being built responsibly on top of it today:

- No sales volume or production-run data: without units sold or produced per model/year, there is no way to weight the specification catalog by actual market demand. Every average in this report is a simple mean across records, not a demand-weighted one.
- Model-level specifications only, no configurator/options detail: the report works from headline trim specifications, not the full options and regional-pricing matrix. Predictive pricing models — especially for a new-model launch — need option-level and market-level granularity to find the price sensitivities that a single "starting price" figure smooths away.

Both gaps are fixable at the source, and neither requires new analytical technique — only a corrected export from BMW's configurator and sales-reporting systems.

11.2 Target Model 1: Launch-Price Forecasting

Goal: recommend a defensible starting price for a new or refreshed model before it reaches market.

- Data required: horsepower, 0-60 time, top speed, body type, drivetrain, and fuel type (all already captured here) plus historical sales volume and competitor pricing for at least 3–5 prior model-year cycles.
- Candidate approach: a multivariate regression predicting starting price from horsepower, acceleration, body type, and drivetrain — this report's Section 7 correlations ($r = 0.87$ for horsepower, $r = -0.75$ for 0-60 time) are exactly the features such a model would start from.
- Output: a recommended launch-price band with a confidence interval, refreshed each time a new trim is specified, surfaced alongside the descriptive catalog view referenced in Section 2 — not a separate report.
- Success metric: forecast error (MAPE) against actual announced launch prices, tracked over time as more model-year data accumulates.

11.3 Target Model 2: Positioning Anomaly Detection

Goal: catch a model whose price has drifted out of line with its specification profile within one catalog refresh, rather than at the next generational review.

- Data required: the same specification fields as above, refreshed each time pricing or a trim spec changes.
- Candidate approach: control-chart style thresholds (e.g. flag any model whose price sits more than 2 standard deviations from the price predicted by its own series' historical price-per-horsepower) are enough to start; this is deliberately simple so it is explainable to a non-technical product reviewer.

- Output: an automated flag routed to the same quality-check layer already in the pipeline (Section 2), not a new parallel process.

11.4 Target Model 3: Electrification-Demand Linkage

Section 6 found the i Series carrying the electrified line-up almost entirely on its own, with horsepower and price broadly comparable to petrol equivalents. The more useful question is a forward-looking one: as more series add electrified variants, does that shift buyers away from the petrol equivalent within the same series, or does it grow the series' total addressable market? Testing this properly needs the sales-volume data described above, split by fuel type, which is not currently captured in this dataset at all — it is the single biggest net-new data requirement in this section.

11.5 Data & Infrastructure Requirements Summary

Requirement	Currently Available?	Needed For
Sales volume / production-run data	No	All three models
Configurator / options-level detail	No — only headline trims	Launch-price forecasting
Competitor pricing benchmark set	No	Launch-price forecasting
Fuel-type-split sales history	No	Electrification-demand linkage
Multi-generation trim history	Partial — some legacy records present	Positioning anomaly detection
Quality-check layer	Yes — already in pipeline	All three models (reuse, don't rebuild)

Table 4. Gap analysis between the current dataset and predictive-analytics readiness.

11.6 Suggested Sequencing

A realistic build order, assuming sales-volume data is sourced first:

- Phase 1 (data): add sales-volume/production-run figures and a competitor pricing benchmark set — no modeling work should start before this.
- Phase 2 (baseline models): control-chart anomaly detection and a simple multivariate price regression, both cheap to build and easy to explain to non-technical stakeholders.
- Phase 3 (refinement): move to configurator-level pricing sensitivity and, only once fuel-type-split sales data exists, the electrification-demand linkage model.
- Phase 4 (productionization): surface all of the above in the same BI layer from Section 2, so predictive output sits next to the descriptive KPIs already in this report rather than in a separate tool.

Happy to build out any of these next, or extend this pipeline to a real multi-generation sales export if one becomes available — the same structure above would apply directly.

Appendix: Methodology Notes & Limitations

- Source: 50-record BMW model specification dataset spanning 12 series and model years 1974–2024, provided as a static snapshot rather than a live configurator/sales feed.
- Fields used: Model, Series, Body Type, Model Year, Engine, Horsepower, Transmission, Drivetrain, 0-60 mph, Top Speed, Fuel Type, Starting Price (USD). No nulls were found in any field.
- All per-series figures in this report (Sections 3–8) are computed directly from the full 50-record dataset — none are approximated or read from a source chart, unlike the visual-only estimates typical of a summary-table-only source.
- "Starting Price" reflects each model's listed base MSRP at the time it was recorded and is not adjusted for inflation across the 1974–2024 span; cross-decade price comparisons (Section 3, Figure 1) should be read as specification history, not real purchasing-power trends.
- This dataset has no sales-volume, production-run, or regional-pricing fields, so every average and correlation in this report is unweighted by actual market demand — a limitation carried through explicitly into Section 11's data-gap analysis.

THANK YOU!

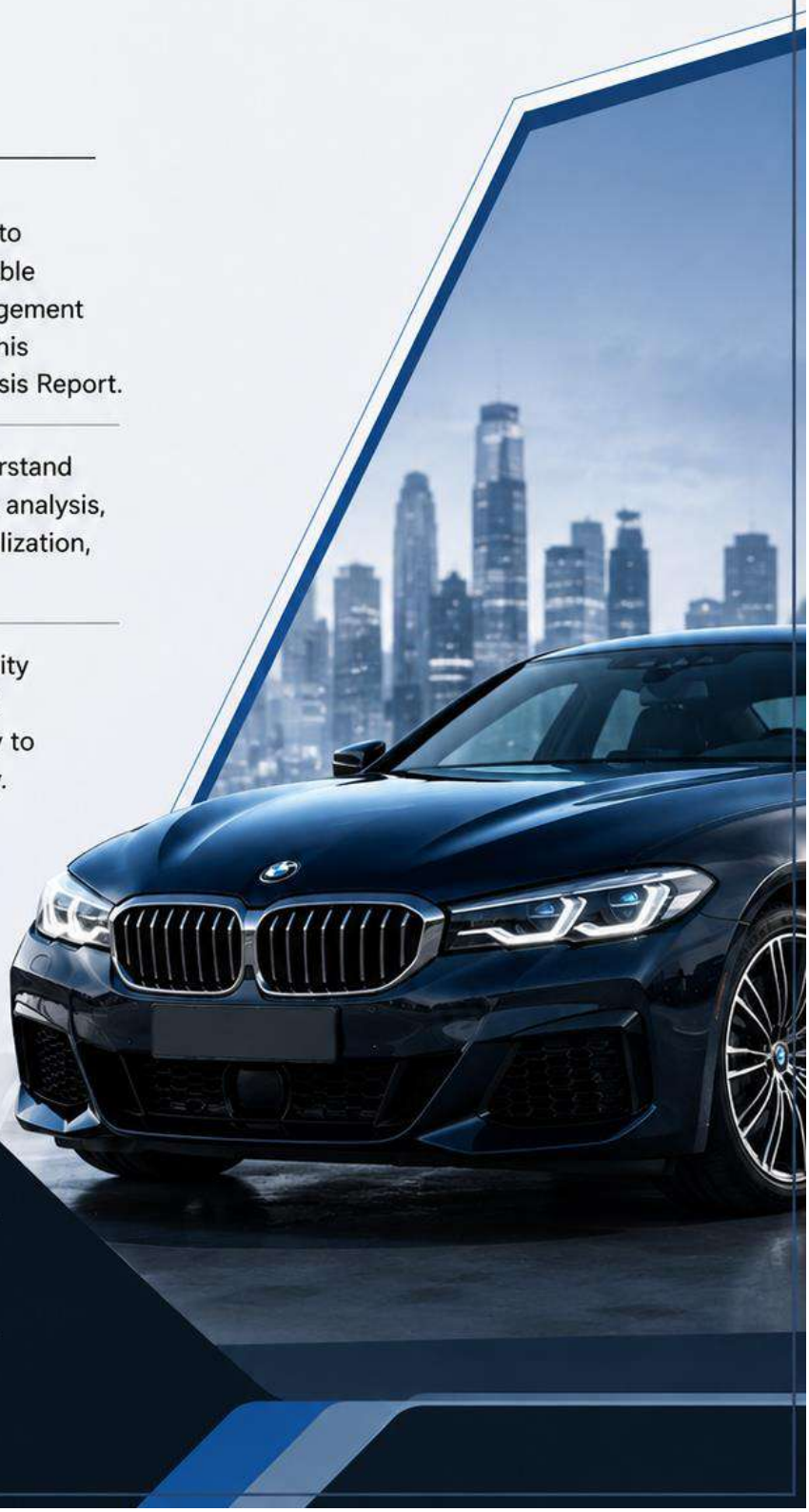
The Ultimate Driving Machine

I sincerely express my gratitude to **Monika Ma'am** for her valuable guidance, support, and encouragement throughout the preparation of this BMW Dataset Descriptive Analysis Report.

This report has helped me understand the practical application of data analysis, descriptive statistics, data visualization, and interpretation of datasets.

I am also thankful to my university for providing the resources and learning environment necessary to complete this work successfully.

Thank You for Your Attention



Submitted by:
Ashish Patel

Submitted to:
Monika Ma'am

University Roll No.:
1250258144